Modulation Strategies for PWM Full Bridge Converter-II

Introduction

In the realm of power electronics, the full-bridge converter stands as a cornerstone for high-power DC-DC conversion, particularly vital in applications like electric vehicle charging infrastructure. As the demand for faster charging and more compact systems grows, engineers are constantly pushing the boundaries of switching frequencies. However, increasing switching frequency often comes at the cost of higher switching losses, which can severely impact efficiency and thermal management. This lecture delves into advanced Pulse Width Modulation (PWM) strategies designed to mitigate these losses by enabling "soft switching" – a technique where switches turn on or off when the voltage across them or current through them is zero.

Understanding these modulation strategies is crucial for designing efficient and reliable power converters. By carefully controlling the turn-on and turn-off sequences of the converter's switches, it's possible to achieve zero-voltage switching (ZVS) or zero-current switching (ZCS), thereby reducing energy dissipation during transitions. This lecture specifically explores two primary types of PWM strategies, Type-1 and Type-2, highlighting their operational differences and their impact on achieving soft switching for the various power semiconductor devices within the full-bridge topology. The insights gained here are fundamental to optimizing converter performance for the demanding requirements of modern power systems.

Core Concepts

Term	Definition	Significance
`Leading Leg`	One half-bridge of the full-bridge converter, typically comprising switches S1 and S2, whose switching transitions are often controlled to initiate the power transfer cycle.	Its switching behavior is critical for establishing the initial voltage across the transformer and can be optimized for soft switching.
`Lagging Leg`	The other half-bridge of the full-bridge converter, typically	Its switching characteristics, particularly ZVS, are often

Term	Definition	Significance
	comprising switches S3 and S4, whose switching transitions often follow the leading leg to complete the power transfer.	more challenging to achieve consistently, especially under varying load conditions.
`Type-1 PWM`	A modulation strategy where the diagonal switches (S1 and S4, or S2 and S3) of the full-bridge converter are turned off simultaneously.	This method simplifies control but typically results in hard turn-on for some switches, leading to higher switching losses.
`Type-2 PWM`	A modulation strategy where the diagonal switches of the full-bridge converter are turned off at different, staggered times.	This strategy aims to facilitate soft switching for more devices, particularly the leading leg switches, by creating a "zero state" period.
`Zero Voltage Switching (ZVS)`	A soft-switching technique where a power semiconductor switch turns on or off when the voltage across it is zero.	ZVS significantly reduces switching losses, improves efficiency, and allows for higher switching frequencies, leading to smaller component sizes.
`Hard Switching`	The conventional method of turning on or off a power semiconductor switch when there is significant voltage across it or current through it.	Hard switching results in substantial switching losses, generates electromagnetic interference (EMI), and limits the maximum operating frequency.
`L_leakage`	The leakage inductance inherent in the primary winding of the high-frequency transformer used in the converter.	This inductance plays a crucial role in the resonant transitions that enable soft switching, particularly in conjunction with device capacitances.

Term	Definition	Significance
`IP current`	The current flowing through the primary winding of the transformer.	It is the primary energy transfer medium and its magnitude and direction are key factors in charging and discharging device capacitances for ZVS.
`IL current`	The current flowing through the output inductor on the secondary side of the converter.	This current, when reflected to the primary side, contributes significantly to the energy available for achieving ZVS in the leading leg.
`Zero State`	A condition in the converter's operation where the voltage across the primary winding of the transformer (VAB) is momentarily zero.	This state is intentionally created in Type-2 PWM to allow for the discharge of device capacitances, facilitating ZVS turn-on for certain switches.

Detailed Analysis

The efficiency and performance of a PWM full-bridge converter are heavily influenced by its modulation strategy, particularly concerning the achievement of soft switching. This section details the two primary types of PWM, Type-1 and Type-2, outlining their operational sequences and the implications for zero-voltage switching (ZVS) of the power semiconductor devices.

1. Type-1 PWM: Simultaneous Diagonal Switch Turn-Off

Type-1 PWM is characterized by the simultaneous turn-off of diagonal switches, such as S1 and S4, or S2 and S3. This strategy corresponds to modulation methods 1 to 3 as previously discussed. The operation unfolds in several distinct phases:

- **Initial Conduction Phase (t < t0):** Switches S1 and S4 are in conduction. The voltage across the transformer primary (VAB) is equal to the input voltage (Vin). The primary current (IP) is a reflection of the output inductor current (IL), which is assumed to be nearly constant due to a large output inductance.

- **Capacitor Charging/Discharging Phase (t0 < t < t1):** At time t0, S1 and S4 are simultaneously turned off by removing their gate pulses. Due to the presence of device output capacitances (C1, C2, C3, C4) across the switches, the IP current now charges and discharges these capacitors.
- C1 charges from 0V, and C2 discharges from Vin.
- C3 discharges from Vin, and C4 charges from 0V.
- This process causes VAB to fall from Vin with a limited slope, determined by the IP current and the combined capacitance (e.g., -IP / (C1+C2) and -IP / (C3+C4)). This limited dv/dt ensures a soft turn-off for S1 and S4.
- **Diode Conduction Phase (t1 < t < t2):** As VAB continues to fall, it eventually reaches -Vin. At this point, the body diodes D2 and D3 (across S2 and S3, respectively) begin to conduct.
- With D2 and D3 conducting, the A terminal is connected to the negative DC link, and the B terminal to the positive DC link, establishing VAB = -Vin.
- The IP current, which was flowing in the positive direction, now starts to decay due to the applied negative voltage across the transformer primary.
- **Resonance and Oscillations Phase (t2 < t < t3):** The IP current continues to decay and eventually reaches zero. At this point, the diodes D2 and D3 cease conduction.
- The transformer's leakage inductance (L_leakage) now resonates with the device capacitances (C1, C2, C3, C4).
- This resonance causes oscillations in both the VAB voltage and the IP current. These oscillations are typically damped by the winding resistance of the transformer.
- The voltage across the switches S2 and S3 (which are about to be turned on) will also oscillate.
- **Switch Turn-On Phase (at t3):** At time t3, gate pulses are applied to S2 and S3 to turn them on.
- Due to the oscillations and the potentially non-zero voltage across S2 and S3 at the instant of turn-on, these switches may experience a "hard turn-on."
- A hard turn-on occurs when a switch turns on with a finite voltage across it, leading to significant switching losses.
- **Summary of Type-1 PWM Soft Switching:** In Type-1 PWM, the switches (S1, S4, S2, S3) achieve zero-voltage turn-off (ZVS turn-off) due to the controlled dv/dt provided by the device capacitances. However, the subsequent turn-on of the complementary switches (S2, S3 or S1, S4) often

results in hard switching due to the resonant oscillations and the inability to guarantee zero voltage at the turn-on instant.

2. Type-2 PWM: Staggered Diagonal Switch Turn-Off

Type-2 PWM employs a staggered turn-off of diagonal switches, meaning they do not turn off simultaneously. This strategy corresponds to modulation methods 4 to 9. This approach aims to improve ZVS conditions, particularly for the turn-on of switches.

- **Initial Conduction Phase (t < t0):** Similar to Type-1, S1 and S4 are initially in conduction, and VAB = Vin. IP current is flowing, reflecting the output current IL.
- **Leading Leg Turn-Off and Zero State Creation (t0 < t < t1):** At t0, only S1 (from the leading leg) is turned off, while S4 (from the lagging leg) remains in conduction.
- The IP current now charges C1 and discharges C2.
- As C2 discharges, the voltage across it (VC2) falls from Vin towards zero. Simultaneously, VC1 rises from zero towards Vin.
- This process causes VAB to transition from Vin towards zero. When VC2 reaches zero, the body diode D2 (across S2) starts to conduct.
- At this point, VAB becomes zero, creating a "zero state." This state is crucial because it allows the voltage across S2 to become zero before it is turned on.
- The energy required for this discharge of C2 and charge of C1 comes from the IP current, which is a reflection of the output inductor current (IL). Since the output inductor (L) is typically large, and it effectively comes in series with the transformer's leakage inductance (L_leakage) during this phase, the combined energy (half * (L_leakage + L/n^2) * IP^2) is usually sufficient to fully discharge C2 and charge C1.
- **Leading Leg Turn-On (at t1):** Once D2 starts conducting (meaning VC2 is zero), a gate pulse is applied to S2.
- Since the voltage across S2 is already zero due to D2 conduction, S2 turns on with zero voltage (ZVS turn-on). This is a significant advantage for the leading leg switches (S1 and S2).
- **Lagging Leg Turn-Off and Transition to -Vin (t1 < t < t2):** After the leading leg transition, the converter is in a zero state (VAB = 0). Now, S4 (from the lagging leg) is turned off. S2 is now on.
- The IP current now charges C4 and discharges C3.

- As C3 discharges, VC3 falls from Vin towards zero, and VC4 rises from zero towards Vin.
- This causes VAB to transition from zero towards -Vin. The dv/dt is limited by the charging/discharging of C3 and C4 by the IP current, ensuring a soft turn-off for S4.
- As VAB reaches -Vin, the body diode D3 (across S3) starts to conduct.
- **Lagging Leg Turn-On (at t2):** A gate pulse is applied to S3 to turn it on.
- Achieving ZVS turn-on for the lagging leg switches (S3 and S4) is more challenging. During the discharge of C3 and charge of C4, the output inductor (L) is effectively short-circuited by the secondary side diodes (DR1 and DR2) and does not contribute its energy to the primary side.
- Therefore, only the energy stored in the leakage inductance (half * L_leakage * IP^2) is available to discharge C3 and charge C4.
- If the load current (and thus IP) is low, or if the device capacitances (C3, C4) are large, this energy may be insufficient to fully discharge C3 to zero before S3 is turned on.
- Consequently, S3 may experience a hard turn-on, especially at light load conditions.
- **Summary of Type-2 PWM Soft Switching:** Type-2 PWM successfully achieves ZVS turn-on for the leading leg switches (S1 and S2) because the large output inductor contributes energy for capacitor discharge. However, ZVS turn-on for the lagging leg switches (S3 and S4) is difficult to guarantee, particularly at light loads, as only the smaller leakage inductance energy is available for the resonant transition. Both leading and lagging leg switches achieve soft turn-off (limited dv/dt).

In conclusion, while both Type-1 and Type-2 PWM offer advantages, Type-2 PWM is generally preferred for its ability to achieve ZVS turn-on for the leading leg, which often carries higher currents. The challenge remains in ensuring ZVS for the lagging leg, especially under varying load conditions, which is an area of ongoing research and design optimization.

Key Takeaways

- [ ] Type-1 PWM involves simultaneous turn-off of diagonal switches (e.g., S1 and S4).
- [ ] Type-1 PWM results in soft turn-off (limited dv/dt) for all switches but typically leads to hard turn-on for the complementary switches due to resonant oscillations.

- [ ] Type-2 PWM involves staggered turn-off of diagonal switches, creating a "zero state" where VAB is momentarily zero.
- [ ] In Type-2 PWM, the leading leg switches (S1, S2) reliably achieve Zero Voltage Switching (ZVS) turn-on, even at low loading conditions.
- [ ] The ZVS turn-on for the leading leg in Type-2 PWM is facilitated by the combined energy of the transformer's leakage inductance and the reflected output inductor.
- [ ] ZVS turn-on for the lagging leg switches (S3, S4) in Type-2 PWM is difficult to achieve, especially at light loads, because only the leakage inductance energy is available for capacitor discharge.
- [ ] Both Type-1 and Type-2 PWM strategies generally ensure soft turn-off (limited dv/dt) for all switches.
- [ ] The choice of PWM strategy impacts the overall efficiency and thermal performance of the full-bridge converter, particularly at high switching frequencies.

Conclusion

The detailed examination of Type-1 and Type-2 PWM strategies reveals the intricate relationship between modulation techniques and the achievement of soft switching in full-bridge converters. While Type-1 offers simplicity in control, its inherent hard turn-on characteristics for certain switches can lead to increased switching losses, limiting its applicability in high-frequency, high-efficiency designs. Type-2 PWM, with its staggered turn-off, presents a more sophisticated approach, successfully enabling zero-voltage turn-on for the critical leading leg switches. This is a significant advantage, as it directly contributes to higher efficiency and reduced thermal stress, which are paramount for compact and reliable power systems.

However, the challenge of consistently achieving ZVS for the lagging leg switches in Type-2 PWM, particularly under light load conditions, underscores the complexities of power converter design. The availability of sufficient energy from the leakage inductance to discharge device capacitances is a critical factor that designers must consider. Ultimately, the selection of an appropriate modulation strategy is a strategic decision, balancing control complexity, component stress, and overall system efficiency. As power demands continue to evolve, a deep understanding of these modulation techniques will remain essential for engineers striving to develop the next generation of high-performance, soft-switched power converters for applications like advanced charging infrastructure.